

INSTITUTE OF ECE
ECA14 – EMBEDDED SYSTEMS
List of Experiments

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STUDY OF PROTEUS AND KEIL MICRO VISION

AIM: To study the working procedures of Proteus and Keil Micro vision software.

Keil Micro Vision is a free software which solves many of the main points for an embedded program developer. This software is an integrated development environment (IDE), which integrated a text editor to write programs, a compiler and it will convert your source code to hex files too. μ Vision4 introduces a flexible window management system, enabling us to drag and drop individual windows anywhere on the visual surface including support for Multiple Monitors.

KEIL PROCEDURE:

1. Open the software, Click on project and open new version project.
2. Create a new project file
3. Enter AT89C51
4. Click NO
5. Click [Ctrl +N] and Type the code
6. Open project and click Build target
7. Open Build target and open source file and ADD, CLOSE
8. Click build target
9. Next debug start and stop
10. Open peripherals and select port 2
11. Now run the program in Debug
12. Open project and click optional properties and in that give output as hex file.
13. Create hex file.

PROTEUS PROCEDURE:

- Open proteus by clicking run as administrator.
- Open new project and enter the file name.
- Click next, next, next and finish.
- Click P symbol and search keyword and place the required components
- Now connect the components as required
- Give input to AT89C51 as HEX file.
- Start the simulation process

RESULT: Thus the Proteus and Keil Micro vision software were studied.

Exp No: 1

FLASHING OF LED USING AT89C51 MICROCONTROLLER USING PROTEUS

AIM:

To Write an assembly language program to flash the LED using AT89C51

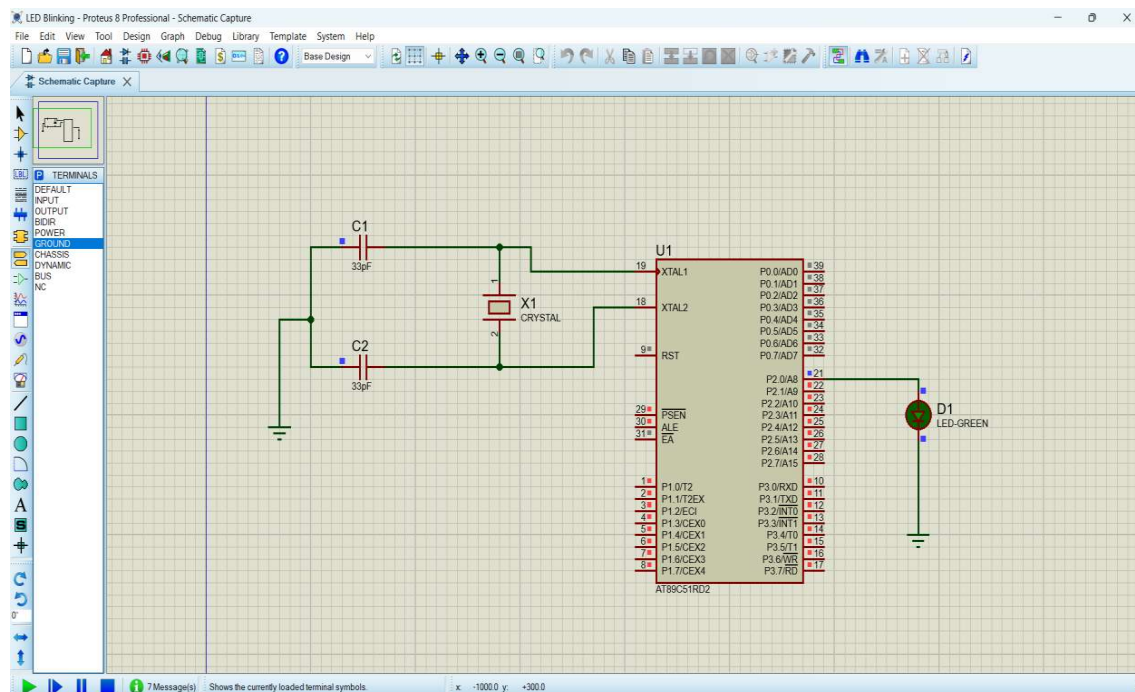
SOFTWARES REQUIRED:

- Proteus software

PROGRAM

```
ORG 0000H
UP: SETB P2.0
  ACALL DELAY
  CLR P2.0
  ACALL DELAY
  SJMP UP
DELAY: MOV R4,#35
      H1:MOV R3,#255
      H2:DJNZ R3,H2
          DJNZ R4,H1
      RET
      END
```

CIRCUIT DIAGRAM:



RESULT

Thus the program has been successfully verified and executed.

Exp No: 2

GENERATION OF SQUARE WAVE USING PROTEUS

AIM:

To write an assembly language program to Generate square wave using AT89C51.

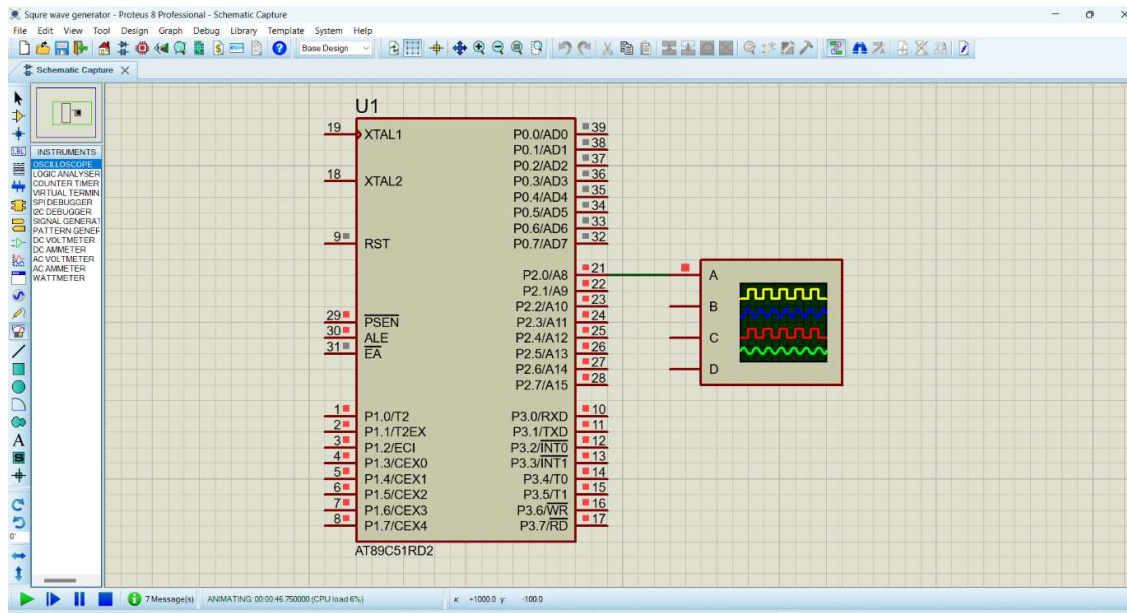
SOFTWARE REQUIRED:

- Proteus 8 software.

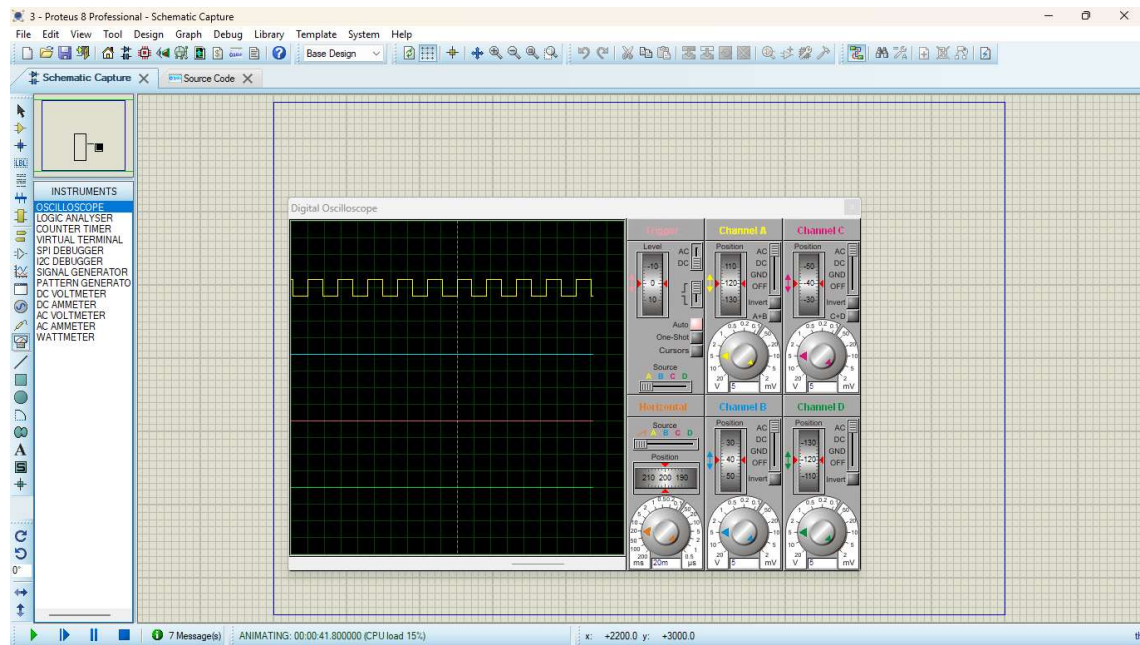
PROGRAM

```
ORG 0000H
UP: SETB P2.0
  ACALL DELAY
  CLR P2.0
  ACALL DELAY
  SJMP UP
DELAY: MOV R4,#35
      H1:MOV R3,#255
      H2:DJNZ R3,H2
          DJNZ R4,H1
      RET
      END
```

CIRCUIT DIAGRAM:



OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.

Exp No: 3

STEPPER MOTOR USING AT89C51 USING PROTEUS

AIM:

To write an assembly language program for Stepper Motor Using AT89C51 using Keil and Proteus

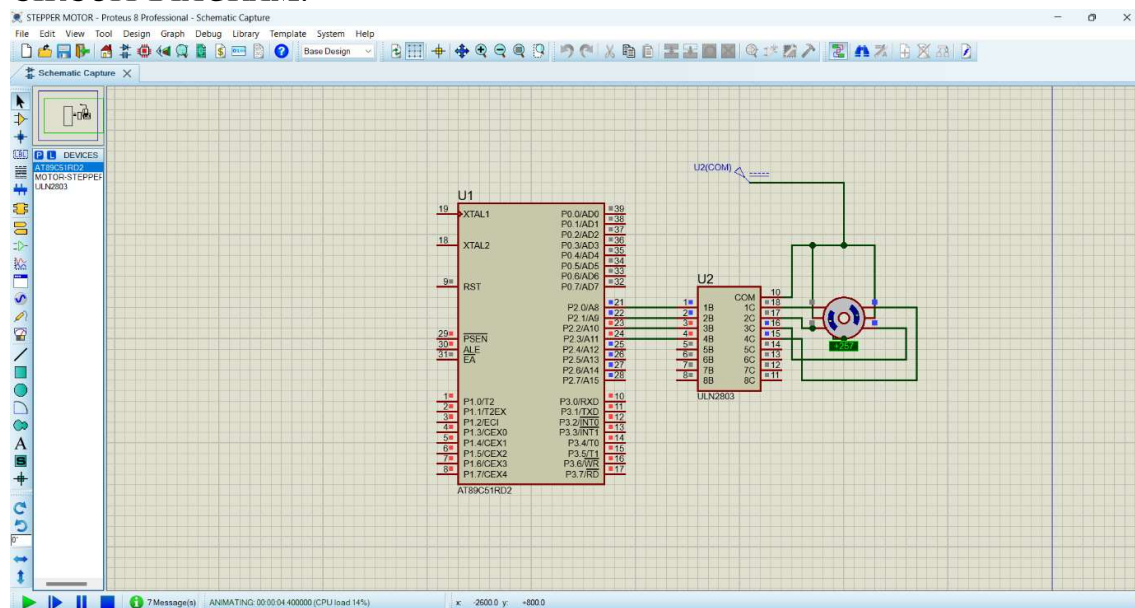
SOFTWARE REQUIRED:

- Proteus 8 software.

PROGRAM:

```
ORG 0000H
UP: MOV P2,#09H
ACALL DELAY
MOV P2,#0CH
ACALL DELAY
MOV P2,#06H
ACALL DELAY
MOV P2,#03H
ACALL DELAY
SJMP UP
DELAY:MOV R4,#18
H1:MOV R3,#255
H2:DJNZ R3,H2
DJNZ R4,H1
RET
END
```

CIRCUIT DIAGRAM:



RESULT:

Thus the program has been successfully verified and executed.

Exp No: 4**7 SEGMENT DISPLAY USING AT89C51 USING PROTEUS****AIM:**

To write an assembly language program for 7 Segment Display Using AT89C51 using Keil and Proteus

SOFTWARE REQUIRED:

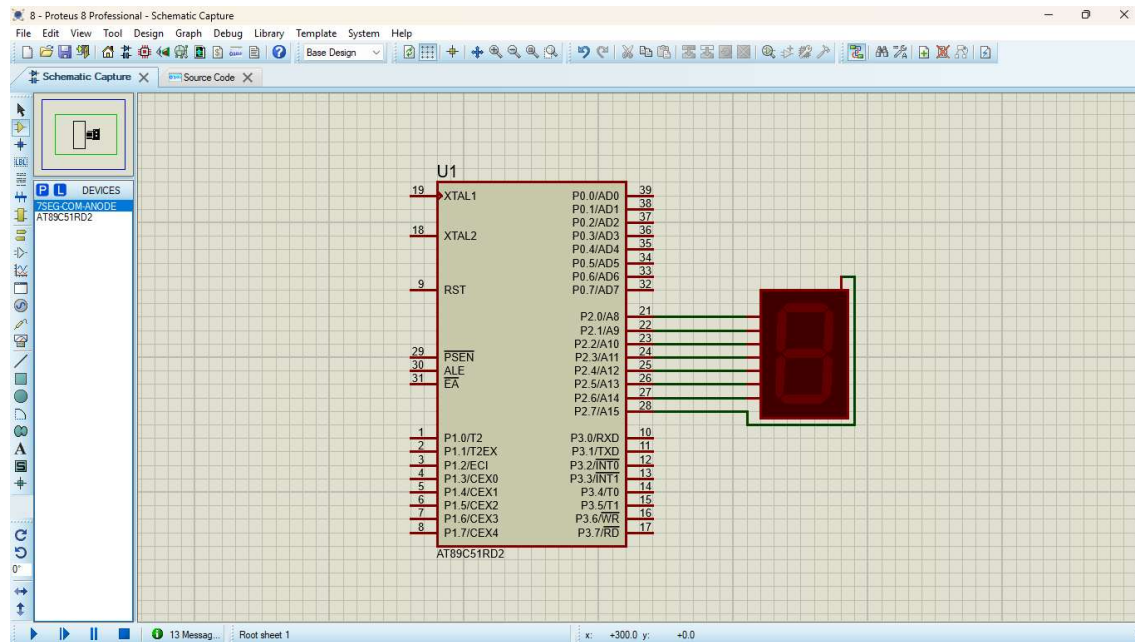
- Proteus 8 software.

PROGRAM:

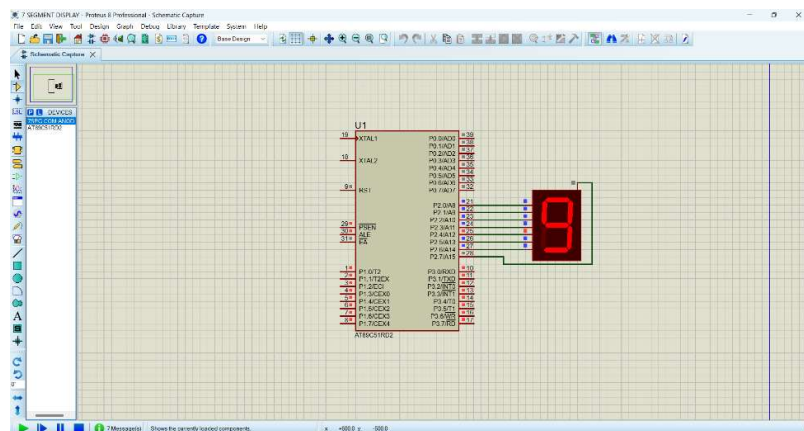
```
ORG 000H
UP:MOV P2,#0C0H
ACALL DELAY
MOV P2,#0F9H
ACALL DELAY
MOV P2,#0A4H
ACALL DELAY
MOV P2,#0B0H
ACALL DELAY
MOV P2,#99H
ACALL DELAY
MOV P2,#92H
ACALL DELAY
MOV P2,#82H
ACALL DELAY
MOV P2,#0F8H
ACALL DELAY
MOV P2, #80H
ACALL DELAY
MOV P2,#90H
ACALL DELAY

DELAY: MOV R5,#10
H1:MOV R4,#180
H2:MOV R3,#255
H3:DJNZ R3,H3
DJNZ R4,H2
DJNZ R5,H1
RET
END
```


CIRCUIT DIAGRAM:



OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.

Exp No: 9

ARDUINO BASED MANUAL ELECTRONIC COUNTER USING PROTEUS

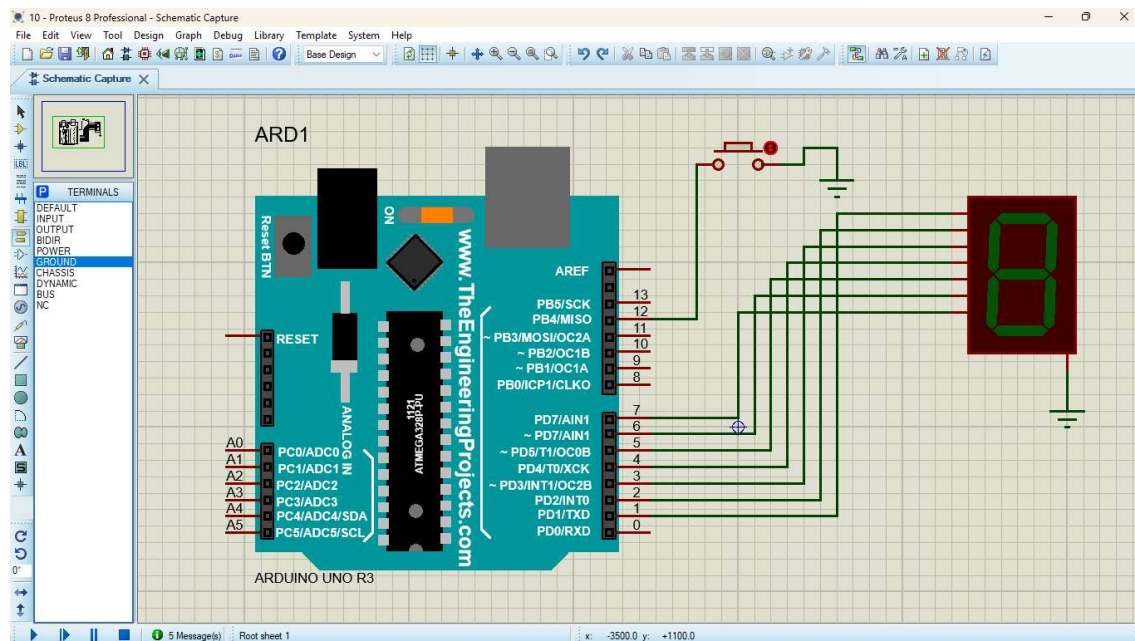
AIM:

To write an Embedded C program for Manual Electronic counter using Arduino IDE and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software
- Arduino IDE

CIRCUIT DIAGRAM:



PROGRAM:

```
int x0,x1,x2,x3,x4,x5,x6,x7,x8,x9;
```

```
int delay_time=200;
```

```
void setup() {
```

```
    //configure pin2 as an input and enable the internal pull-up resistor
```

```
    pinMode(12, INPUT_PULLUP);
```

```
    pinMode(1,OUTPUT);
```

```

pinMode(2,OUTPUT);
pinMode(3,OUTPUT);
pinMode(4,OUTPUT);
pinMode(5,OUTPUT);
pinMode(6,OUTPUT);
pinMode(7,OUTPUT);

}

void loop() {
  //read the pushbutton value into a variable
  int sensorVal = digitalRead(12);

  if (sensorVal == LOW) {
    x0=true;
  }
  while(x0){
    zero();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x1=true;
      x0=false;
    }
  }
  while(x1){
    one();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
      x2=true;
      x1=false;
    }
  }
}

```

```
    }  
while(x2){  
    two();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x3=true;  
        x2=false;  
    }  
}  
while(x3){  
    three();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x4=true;  
        x3=false;  
    }  
}  
while(x4){  
    four();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x5=true;  
        x4=false;  
    }  
}  
while(x5){  
    five();  
    sensorVal = digitalRead(12);  
    if (sensorVal == LOW) {  
        x6=true;
```

```
        x5=false;
    }
}
while(x6){
    six();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x7=true;
        x6=false;
    }
}
while(x7){
    seven();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x8=true;
        x7=false;
    }
}
while(x8){
    eight();
    sensorVal = digitalRead(12);
    if (sensorVal == LOW) {
        x9=true;
        x8=false;
    }
}
while(x9){
    nine();
    sensorVal = digitalRead(12);
```

```
        if (sensorVal == LOW) {  
            x0=true;  
            x9=false;  
        }  
    }  
  
}
```

```
void zero()  
{  
    for(int i=1;i<7;i++)  
    {  
        digitalWrite(i,HIGH);  
        digitalWrite(7,LOW);  
    }  
    delay(delay_time);  
}
```

```
void one()  
{  
    digitalWrite(1,LOW);  
    digitalWrite(2,HIGH);  
    digitalWrite(3,HIGH);  
    digitalWrite(4,LOW);  
    digitalWrite(5,LOW);  
    digitalWrite(6,LOW);  
    digitalWrite(7,LOW);  
    delay(delay_time);  
}
```

```
void two()
{
    digitalWrite(1,HIGH);
    digitalWrite(2,HIGH);
    digitalWrite(3,LOW);
    digitalWrite(4,HIGH);
    digitalWrite(5,HIGH);
    digitalWrite(6,LOW);
    digitalWrite(7,HIGH);
    delay(delay_time);
}
```

```
void three()
{
    digitalWrite(1,HIGH);
    digitalWrite(2,HIGH);
    digitalWrite(3,HIGH);
    digitalWrite(4,HIGH);
    digitalWrite(5,LOW);
    digitalWrite(6,LOW);
    digitalWrite(7,HIGH);
    delay(delay_time);
}
```

```
void four()
{
    digitalWrite(1,LOW);
    digitalWrite(2,HIGH);
    digitalWrite(3,HIGH);
    digitalWrite(4,LOW);
```

```
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void five()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,LOW);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```

```
void six()  
{  
digitalWrite(1,HIGH);  
digitalWrite(2,LOW);  
digitalWrite(3,HIGH);  
digitalWrite(4,HIGH);  
digitalWrite(5,HIGH);  
digitalWrite(6,HIGH);  
digitalWrite(7,HIGH);  
delay(delay_time);  
}
```



```
void seven()
{
    digitalWrite(1,HIGH);
    digitalWrite(2,HIGH);
    digitalWrite(3,HIGH);
    digitalWrite(4,LOW);
    digitalWrite(5,LOW);
    digitalWrite(6,LOW);
    digitalWrite(7,LOW);
    delay(delay_time);
}
```

```
void eight()
{
    digitalWrite(1,HIGH);
    digitalWrite(2,HIGH);
    digitalWrite(3,HIGH);
    digitalWrite(4,HIGH);
    digitalWrite(5,HIGH);
    digitalWrite(6,HIGH);
    digitalWrite(7,HIGH);
    delay(delay_time);
}
```

```
void nine()
{
    digitalWrite(1,HIGH);
    digitalWrite(2,HIGH);
    digitalWrite(3,HIGH);
    digitalWrite(4,HIGH);
```

```

digitalWrite(5,LOW);

digitalWrite(6,HIGH);

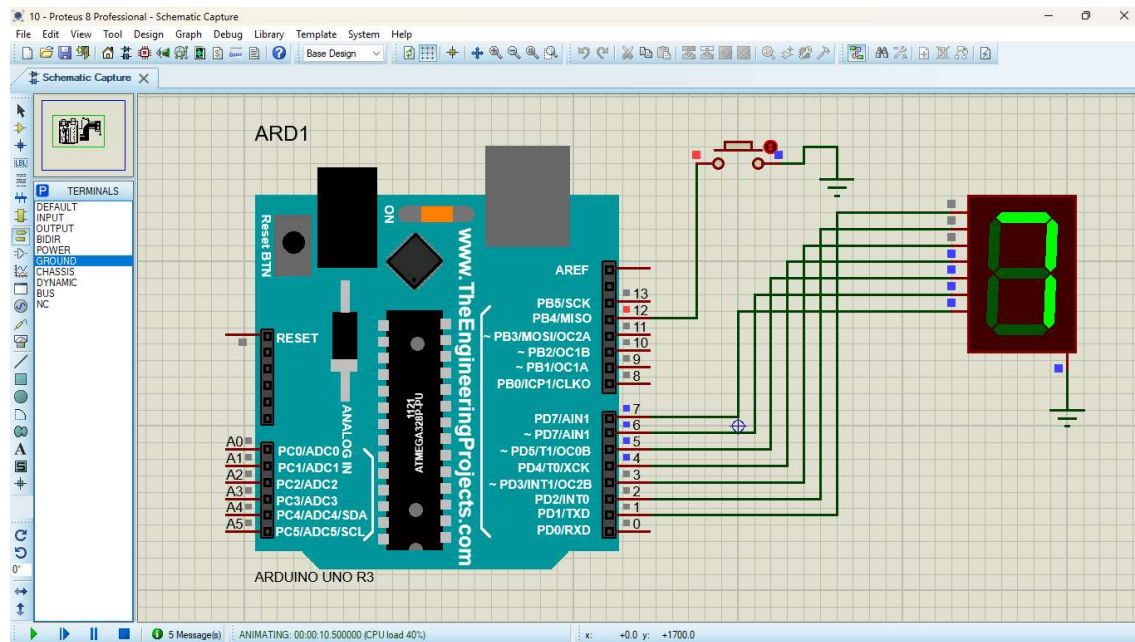
digitalWrite(7,HIGH);

delay(delay_time);

}

```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.

Exp No: 6

ARDUINO TO 16x2 LCD DISPLAY USING PROTEUS

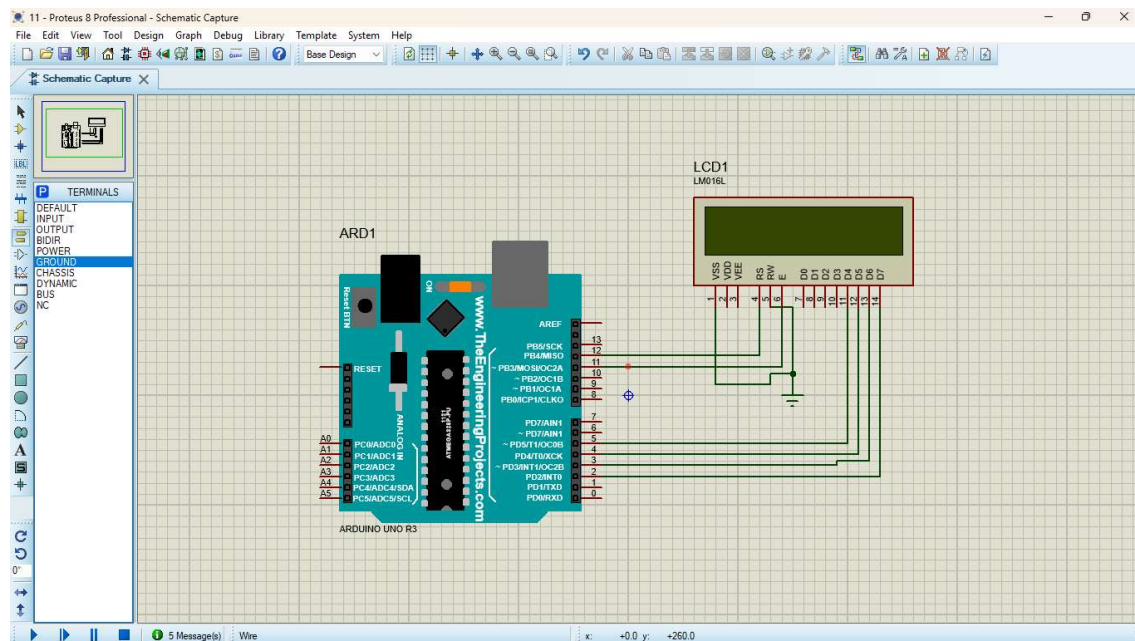
AIM:

To write an Embedded C program to interface Arduino Uno and 16 x 2 LCD display using Arduino Uno and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software.
- Arduino IDE software

CIRCUIT DIAGRAM:



PROGRAM:

Arduino to 16x2 LCD Display using Proteus

```
#include <LiquidCrystal.h>
```

```
LiquidCrystal lcd (7, 8, 9, 10, 11, 12);
```

```
// your PWM numbers may be different
```

```
void setup() {
```

```
  lcd.begin(16, 2);
```

```
lcd.print("put your message here");
```

```
}
```

```
void loop () {
```

```
for (int i=0; i<13; i++)
```

```
}
```

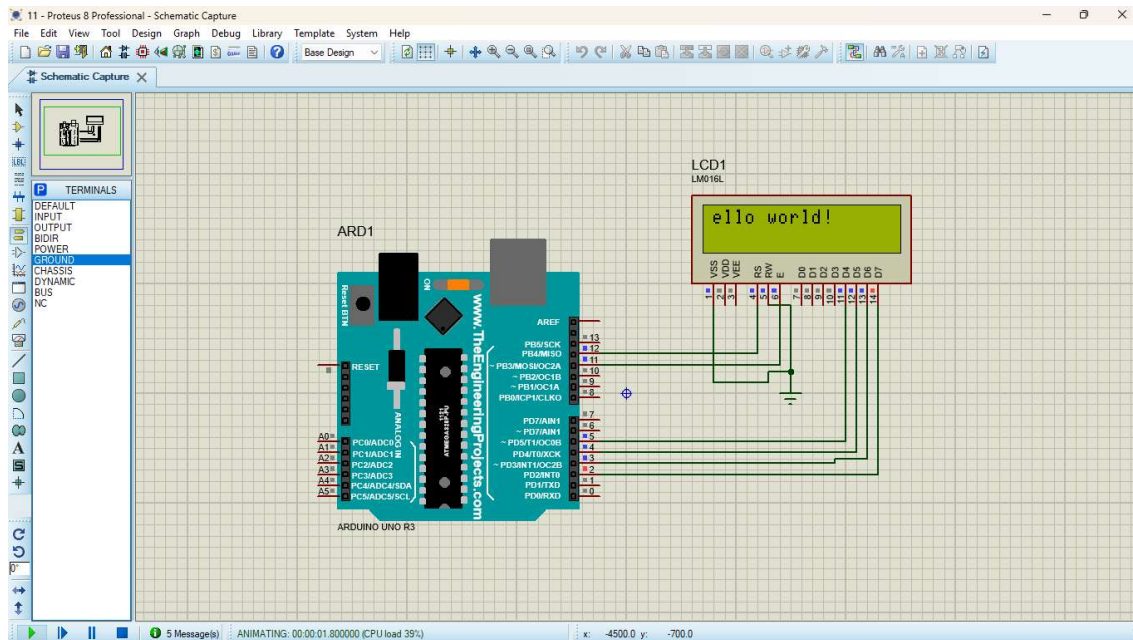
```
lcd.scrollDisplayLeft();
```

```
delay(700);
```

```
}
```

```
}
```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.

Exp No: 7

SERIAL COMMUNICATION USING ARDUINO WITH PROTEUS

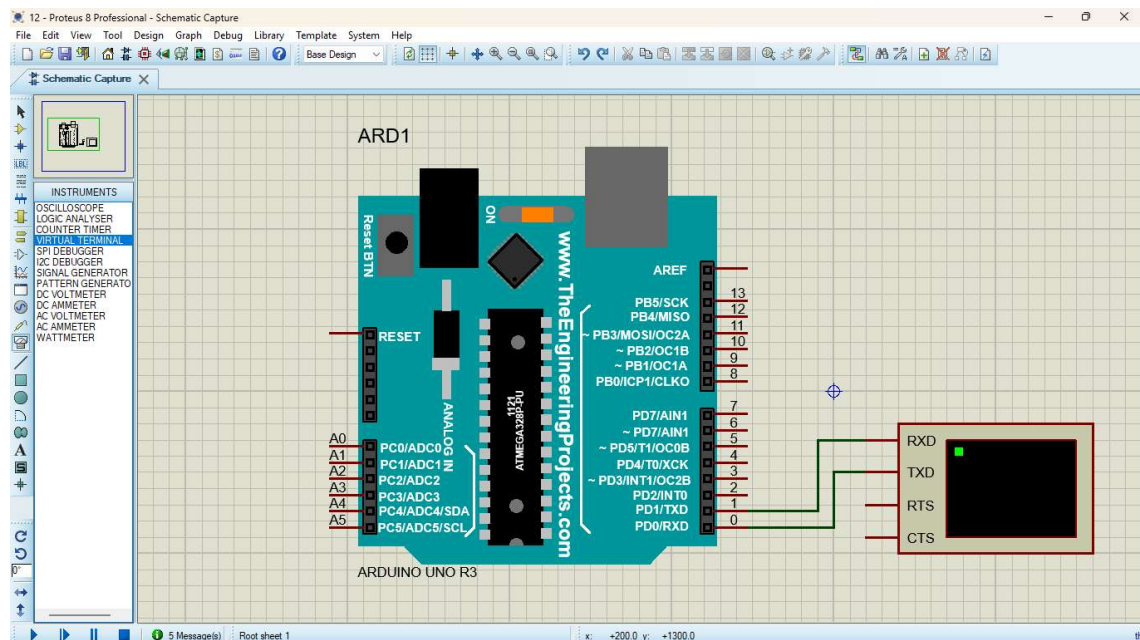
AIM:

To write an Embedded C program for serial communication using Arduino Uno and Proteus

SOFTWARE REQUIRED:

- Proteus 8 software.
- Arduino IDE software

CIRCUIT DIAGRAM:



PROGRAM:

```
void setup()
{
  Serial.begin(9600); // send and receive at 9600 baud
}

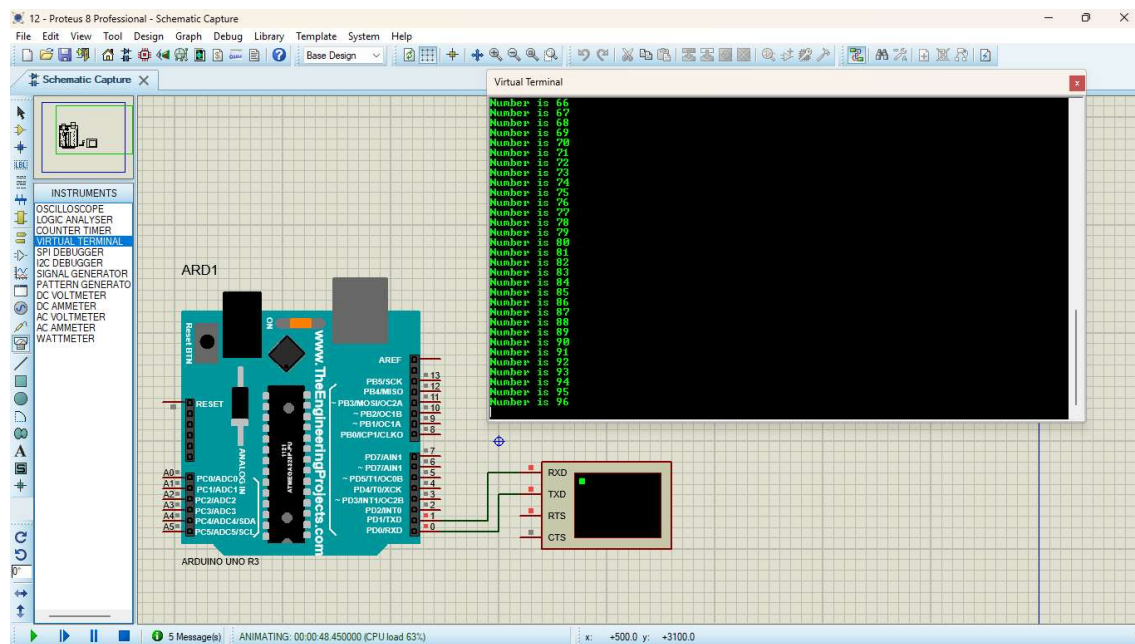
int number = 0;
```

```

void loop()
{
    Serial.print("Number is ");
    Serial.println(number); // print the number
    delay(500); // delay half second between numbers
    number++; // to the next number
}

```

OUTPUT:



RESULT:

Thus the program has been successfully verified and executed.